CSCI 332, Fall 2025

Homework 1

Due Monday, September 1 Anywhere on Earth (6am Tuesday)

Submission Requirements

- Type or clearly hand-write your solutions into a PDF format so that they are legible and professional. Submit your PDF on Gradescope.
- Do not submit your first draft. Type or clearly re-write your solutions for your final submission. If your submission is not legible, we will ask you to resubmit.
- Use Gradescope to assign problems to the correct page(s) in your solution. If you do not do this correctly, we will ask you to resubmit.

Academic Integrity

Remember, you may access ***any*** resource in preparing your solution to the homework. However, you ***must***

- write your solutions in your own words, and
- credit every resource you use (for example: “Bob Smith helped me on this problem. He took this course at UM in Fall 2020”; “I found a solution to a problem similar to this one in the lecture notes for a different course, found at this link: www.profzeno.com/agreatclass/lecture10”; “I asked ChatGPT how to solve part (c)"; “I put my solution for part (c) into ChatGPT to check that it was correct and it caught a missing case.”) If you use the provided LaTeX template, you can use the `sources` environment for this. Ask if you need help!

1. (0 points) List the resources you used for each problem in this homework. If you used no resources except lecture notes and videos and the linked textbook, you can say "none". If you used an AI tool, you should put a link to your conversation here.

   You will be asked to resubmit your homework if this section is blank.

   Remember that no matter what resources you consulted, you must write your solutions *in your own words*.

2. For your upcoming family reunion, you are in charge of one of the day's activities. You decide to have the family head out in two-person canoes. However, you know that you need to be careful in pairing your family members up, because they can get jealous if they get stuck spending time with one person when they would prefer to spend time with another. Because you are taking CSCI 332, you wonder if you could re-work the stable matching definition from the hospital-student problem to work for your canoe pair problem. You consider asking each family member to order the rest of the family members according to their preference for a canoe partner.

   For example, for $n = 4$ family reunion attendees, you might get the following list:

   Alice: Ben, Cleo, Desiree
   Ben: Cleo, Alice, Desiree
   Cleo: Alice, Ben, Desiree
   Desiree: Alice, Ben, Cleo

   (a) (3 points) Re-work the definition of stable matching from the hospital-medical student version of the problem to the canoe pair problem. You should describe the input of the problem and the desired output of the problem.

   (b) (3 points) Does a stable matching always exist for the canoe pair problem?

   (c) (3 points) How do you know?

3. (1 point) Back to the hospital-student stable matching problem: describe and analyze an efficient algorithm to determine whether a given set of preferences has a *unique* stable matching.